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WIRELESS REMOTE DATA MONITORING SYSTEM With TFT LCD Monitor

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Remote monitoring plays a pivotal role in advancing industrial automation, Industry 4.0, and the Industrial Internet of Things (IIoT). A remote data monitoring system (RDMS) continuously tracks parameters such as temperature, pressure, humidity, flow, and machine performance real-time.

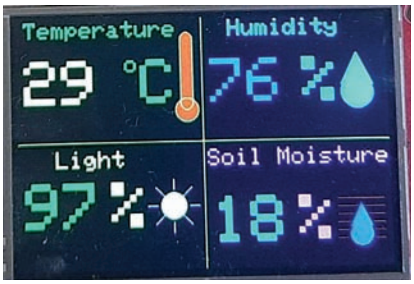

Fig. 1: TFT LCD monitor showing sensor data

By collecting data from strategically placed sensors throughout an industrial plant and transmitting it to a centralised hub, this system enhances operational efficiency, reduces downtime, and ensures timely maintenance, thereby boosting overall productivity and safety.

An RDMS proves especially advantageous in large-scale or hazardous environments where manual monitoring is either impractical or risky. It exemplifies the capabilities of modern technology, enabling smarter, safer, and more efficient industrial processes.

A notable example of an RDMS is found in agriculture. Soil moisture, pH, and humidity sensors installed across farms monitor environmental conditions to promote optimal crop health.

The RDMS presented here is a long-range wireless system with a 10.92cm TFT LCD monitor that displays real-time data from three types of sensors using multicolour graphics and animations. It tracks four environmental parameters: temperature, humidity, ambient light, and soil moisture, and features an impressive range of several kilometres.

As illustrated in Fig. 1, the TFT

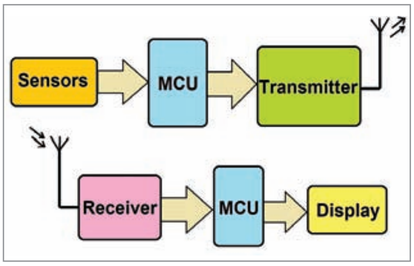

Fig. 2: Block diagram of a wireless RDMS

TABLE 1
BILL OF MATERIALS

Components	Quantity
Arduino Uno/Nano (MOD1, MOD4)	2
Temperature and humidity sensor module DHT11 (S1)	1
Soil moisture sensor module (S3)	1
Ambient light sensor LDR (S2)	1
HC12 module (MOD 2, MOD3)	2
Resistors 1-kilo-ohm (R1-R6)	6
Resistors 2.2-kilo-ohm (R7-R12)	6
Resistors 10-kilo-ohm (R13)	1
10.92cm (4.3-inch) SPI TFT LCD monitor	1
5V regulated power supply	2 (for transmitter and receiver)
Jumper wires and wires	Based on requirement

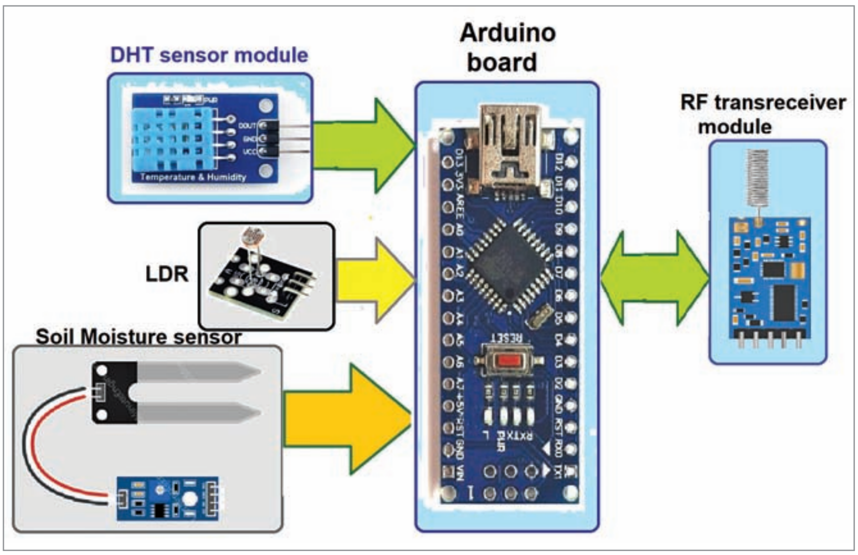

Fig. 3: Block diagram of the remote data transmitter